

ShopEasy Term Project - Reflection Report

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1. Bugs Found by Each Technique

In this project, I didn't encounter any real defects in the starter code—after writing the tests, everything passed. Still, each testing technique would catch different types of mistakes if the code were faulty.

Specification-based testing (Task 1): Testing PriceCalculator against specifications is useful for catching formula errors. For example, if the discount and tax steps were swapped, tests like testTypical108 (100, 10, 20 → 108) would fail. Invalid partition tests also revealed that the calculator accepted negative base prices before Task 3. After adding assertions, these invalid inputs were caught as AssertionError by the contract tests.

Structural testing with JaCoCo (Task 2): This helped identify which branches in ShoppingCart were not covered. Initially, I didn't test the case where a product exists in the cart but has the wrong ID in updateQuantity. JaCoCo highlighted this missing branch, so I added updateQtyWrongId and achieved 100% branch coverage. Structural testing doesn't always find logic bugs but is effective for revealing missing tests.

Design by Contract (Task 3): Adding assertions to addItem and calculate documents the assumptions and ensures illegal states fail fast. Reviewers can see which inputs are valid without examining every test.

Property-based testing with jqwik (Task 4): Properties like monotonicity (higher discount should never increase the total price) would catch errors like a sign mistake in the discount formula across hundreds of input combinations, not just a few hand-picked examples. In my run, jqwik didn't report any failures.

Mocks (Task 5): Using mocks for OrderProcessor let me test behavior without relying on a real warehouse or payment API. For instance, I verified that when isAvailable returns false, charge is never called. If the payment logic were triggered before the inventory check, the inventoryFail test would fail. However, mocks do not test the correctness of the real InventoryService implementation.

2. Most Effective Technique per Unit of Effort

For me, specification-based testing combined with structural coverage was the most efficient. Task 1 helped me understand the price formula and write clear examples. Task 2 then showed gaps in ShoppingCart with JaCoCo, which I could fix with just one or two additional tests.

Property-based testing was powerful but took more time to learn (jqwik annotations, Assume.that, @Provide). Mockito was very helpful for testing OrderProcessor, but it only applied to that one class with external dependencies.

If I had limited time, I would prioritize: specification-based tests and branch coverage first, then contracts, followed by mocks, and finally property-based testing.

3. Testing Pyramid

Most of my test suite is unit tests at the bottom of the testing pyramid: fast tests for PriceCalculator, ShoppingCart, and OrderProcessor (with mocks). There are no integration tests involving a real database or HTTP API, and no end-to-end UI tests.

What's missing are integration tests connecting real components (for example, a real InventoryService with OrderProcessor) and a small set of system tests for the full checkout flow. I also did not implement performance or load tests.

4. If I Had One More Week

I would focus on integration tests for a full checkout process: using a real cart, a simple in-memory inventory, and a fake payment gateway that records charges. This would ensure mocks didn't hide any wiring issues.

I would also run PIT mutation testing on ShoppingCart (as an optional bonus) and add tests for surviving mutants. Finally, I would test edge cases in OrderProcessor that I skipped: empty customerId, null cart, and a cart with many items to verify the inventory loop works correctly.

5. Trade-offs of Mockito

Mocking allowed me to test the decision logic of OrderProcessor in isolation, for example: returning null when stock fails, calling charge only when all items pass, and returning null if payment fails.

However, I couldn't test that a real payment provider actually charged the correct amount or that the database stock was updated. Mocking is less suitable when interactions between real components are important or when mocks mimic the real implementation so closely that the test loses meaning.